

WEN-HORNG SHEU

Phone: (530) 979-6045

Email: wsheu@ucdavis.edu

Links: Personal Website [↗](#), LinkedIn [↗](#)

EDUCATION

PhD in Computer Science at the University of California, Davis; GPA: 4.0/4.0 *2023-Present*

· Research area: Distributed Algorithms, Streaming Algorithms

Master of Computer Science at National Tsing Hua University; GPA: 3.9/4.0 *2019-2021*

· Specialized courses: Approximation Algorithms, Computational Geometry

Bachelor of Computer Science at National Tsing Hua University; GPA: 3.85/4.0 *2015-2019*

· Specialized courses: Advanced Data Structure, Randomized Algorithms, Parallel Algorithm Design

PUBLICATIONS

Following the convention in theoretical computer science, author names are ordered alphabetically unless stated otherwise.

7. **Fixed-Threshold Peeling in Sublinear MPC: Round-Approximation Tradeoffs and Applications**
with Slobodan Mitrović and Theodore Pan
to appear in DISC 2026 (International Symposium on Distributed Computing)
6. **A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic**
with Slobodan Mitrović
SPAA 2025 (ACM Symposium on Parallelism in Algorithms and Architectures)
5. **Faster MPC Algorithms for Approximate Allocation and Matching in Uniformly Sparse Graphs**
with Jakub Łącki, Slobodan Mitrović, and Srikanth Ramachandran
SPAA 2025 (ACM Symposium on Parallelism in Algorithms and Architectures)
4. **Faster Semi-streaming Matchings via Alternating Trees**
with Slobodan Mitrović, Anish Mukherjee, and Piotr Sankowski
ICALP 2025 (EATCS International Colloquium on Automata, Languages, and Programming)
3. **Kernelization and Approximation Algorithms for Finding a Perfect Phylogeny from Mixed Tumor Samples**
Wen-Horng Sheu and Biing-Feng Wang (contribution order)
TCBB 2025 (IEEE Transactions on Computational Biology and Bioinformatics)
2. **New Algorithms for Constructing Frequency Difference Consensus Trees**
Biing-Feng Wang, Chih-Yu Li, and Wen-Horng Sheu (contribution order)
TCBB 2025 (IEEE Transactions on Computational Biology and Bioinformatics)
1. **Parameterized Complexity for Finding a Perfect Phylogeny from Mixed Tumor Samples**
Wen-Horng Sheu and Biing-Feng Wang (contribution order)
SIDMA 2023 (SIAM Journal on Discrete Mathematics)

TALKS AND PRESENTATIONS

- **A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic**
INSAIT, May 2025
- **Toward Optimal Semi-streaming Algorithm for $(1 + \epsilon)$ -approximate Maximum Matching**
Simons Institute for the Theory of Computing, UC Berkeley, Aug 2024
in Workshop on Local Algorithms 2024

RESEARCH EXPERIENCE

Graduate Research Assistant at the University of California, Davis 2023-Present

- Research area: distributed algorithms, streaming algorithms.
- Studied the maximum matching problem in distributed and streaming settings.

Research Assistant at National Tsing Hua University 2021-2023

- Research area: parameterized algorithms, computational biology.
- Proposed new algorithms for problems that have applications in cancer genomics and phylogenetic analysis.
- Created problems for the International Collegiate Programming Contest (ICPC).

TEACHING EXPERIENCE

Teaching Assistant at the University of California, Davis 2024-Present

- Taught Algorithm Design and Analysis (2024, 2025, 2026).
- Taught Special Topics in Theoretical Computer Science (2024).

Teaching Assistant at National Tsing Hua University 2019-2022

- Taught Computational Geometry (2020, 2022).
- Taught Parallel Algorithm Design (2019, 2022).
- Taught Design and Analysis of Algorithms (2019, 2020, 2021).

HONORS AND AWARDS

- **Gold Award** in *the 2019 ICPC Asia Pacific Taipei-Hsinchu Regional Contest*
 - Attended several programming contests, including ICPC, with other students in undergraduate years.
 - Built strong abilities in teamwork and problem-solving.
- **Second Place Award** in *the ACM TAU 2018 Contest on Path Reporting*
 - Undergraduate research project: created a tool to perform timing analysis on input circuits.
- **Grandmaster on Codeforces**
 - Ranked as a grandmaster, within top 1% globally on the online competitive programming platform Codeforces (as of Feb 2021).
 - Placed top 1% (out of 10,000+ contestants globally) in four online programming contests on Codeforces.
- **Google Code Jam 2021 Round 3 Qualifier**
 - Competed in the programming contest Google Code Jam 2021.
 - Placed 255th in Round 3, within top 1% of all 37,000+ participants of the qualification rounds.

ACADEMIC SERVICES

- **Shadow Program Committee Member** for SPAA 2026.
- **External reviewer** for the following conferences and journals.

- Conferences: ICALP (2025), ISAAC (2026), SODA (2025, 2026), SOSA (2025).
- Journals: Distributed Computing (2025), TALG (2025).

SKILLS

Programming Languages	C, C++, Python
Tools	Git, L ^A T _E X
Languages	English (fluent), Chinese (native)